RUTHVIK REDDY ANTHIREDDIGARI

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OBJECTIVE

M.S. Electrical Engineering student specializing in **Analog/Mixed-Signal IC Design** with hands-on transistor-level CMOS design and semiconductor silicon-validation experience. Experienced in Cadence Virtuoso/Spectre, OTAs/op-amps, LDOs, feedback/stability, PVT characterization, noise and distortion analysis. Seeking **Spring 2027 Co-op** or **NCG 2027** Analog/Mixed-Signal IC Design opportunities.

EDUCATION

Arizona State University, Tempe, AZ, USA Aug 2025 - Aug 2027
***Master of Science and Engineering*,** Electrical Engineering.
Relevant coursework: *Analog Integrated Circuits, Advanced Analog Integrated Circuits, VLSI Design, Digital Verification & Testing.*
CGPA: 3.94/4.00

Manipal Institute of Technology, MAHE, KA, IND Oct 2020 - July 2024
Bachelor's in Technology, *Major in Electrical and Electronics Engineering.*

SKILLS

- **Analog IC Design:** CMOS analog circuits, current mirrors, differential amplifiers, OTAs, op-amps, cascode amplifiers, LDOs
- **EDA Tools**: Cadence Virtuoso (Schematic & Layout), Spectre, HSPICE, LTspice, MATLAB
- **Circuit Analysis:** DC, AC & transient analysis, frequency response, stability, CMRR/PSRR, noise & distortion analysis
- **Programming:** Python, C, Verilog, SystemVerilog, VHDL
- **Lab Tools:** Oscilloscopes, spectrum analyzers, function generators, signal generators

INTERNSHIP EXPERIENCE

Renesas Electronics America Inc., Tempe, AZ May 2026 – Aug 2026
Validation Intern – DDR5 Silicon Validation / Memory Interface Team

- Performed post-silicon validation and electrical characterization of **DDR5 data-buffer devices**, supporting high-speed timing measurements and silicon debug across multiple devices and bench setups.
- Developed **Python/SCPI automation** for 4-channel Keysight oscilloscope deskew, reducing inter-channel timing **skew from ~1.04 ps to ~50 fs** for repeatable high-speed measurements.
- Automated DQS-to-BCK **timing characterization**, including differential-channel configuration, triggering, burst-mode acquisition, and min/mean/max/range measurement reporting.
- **Debugged DDR5 validation workflows** using MR/CSR read-write operations, FIFO testing, waveform analysis, and Python test scripts, integrating automation updates through Git/GitLab merge requests.

Delta IOT Solutions Private limited, Hyderabad, India Mar 2025 – Jun 2025
Intern – Embedded

- Developed embedded code **for Arduino/ESP32**-based IoT applications, enabling stable sensor data transfer.
- Designed **2 custom PCBs Ki-Cad**, reducing prototyping debug cycles by 25%.

Research Centre Imarat, DRDO , Hyderabad, India Feb 2024 – June 2024
Intern - Student/Project.

- Engineered a low-latency, power-efficient video data streaming system using **AXI4-Stream and AMBA protocols**, achieving 15% latency reduction and lowering power consumption by 10%.
- Verified RTL designs in VHDL/Verilog with C testbenches; performed synthesis and timing validation using Vivado/Vitis

ACADEMIC PROJECTS

Folded Cascode Amplifier with Class-AB Output (Cadence, CMOS) Spring 2026

- Designed a folded-cascode/Class-AB amplifier targeting high gain and stability; performed transistor sizing, biasing, AC/transient and distortion analysis, achieving ~90 dB gain, >10 MHz UGF, >60° PM, ~10 V/µs SR and ~−50 dB HD3

β-Multiplier Constant-g_m Current Reference Spring 2026

- Designed a **~10 µA β-multiplier reference** using **K=2 device scaling, resistor current-setting and startup circuitry**; verified operation across **2–3 V VDD and −20°C to +85°C**, achieving **<5% $g_m$variation**

Telescopic-Cascode Differential OTA | Cadence/Spectre Spring 2026

- Designed and biased a telescopic-cascode OTA targeting >50 dB gain and ~50 MHz UGF; characterized gain/phase, CMRR/PSRR and common-mode operation, achieving ~90 dB CMRR.

Low-Dropout (LDO) Regulator Design & OTA (Cadence, 2.5V CMOS) Fall 2025

- Designed a **2.3 V LDO for 0.1–50 mA loads** using an **OTA error amplifier, PMOS pass device and resistive feedback**; evaluated DC, loop stability and load transients, achieving **~90° PM and stable 2.3 V regulation**

CERTIFICATIONS

- VLSI Design Flow : RTL to GDS, NPTEL – IIIT, Delhi (*Link*)